

MACHINEPULSE · DESIGN PRECIS

MachinePulse design precis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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Source	docs/02-concept.md @ 2616d17		

REVISION HISTORY

Version	Date	Author	Change
0.4	2026-09-25	Amish Chadha	Recommendations accepted by Amish (DDR-002)
0.3	2026-09-25	Amish Chadha	TRL 3 update; design choices adopted per MPL-DDR-001; numbers from MPL-CAL-001; stand-offs, CT clamp, 1.50 V bias, +-16 g range, module variant and 3 mm pads added; GA drawing MPL-DWG-001
0.2	2026-09-25	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, safety, media)
0.1	2026-09-25	Amish Chadha	Initial scaffold

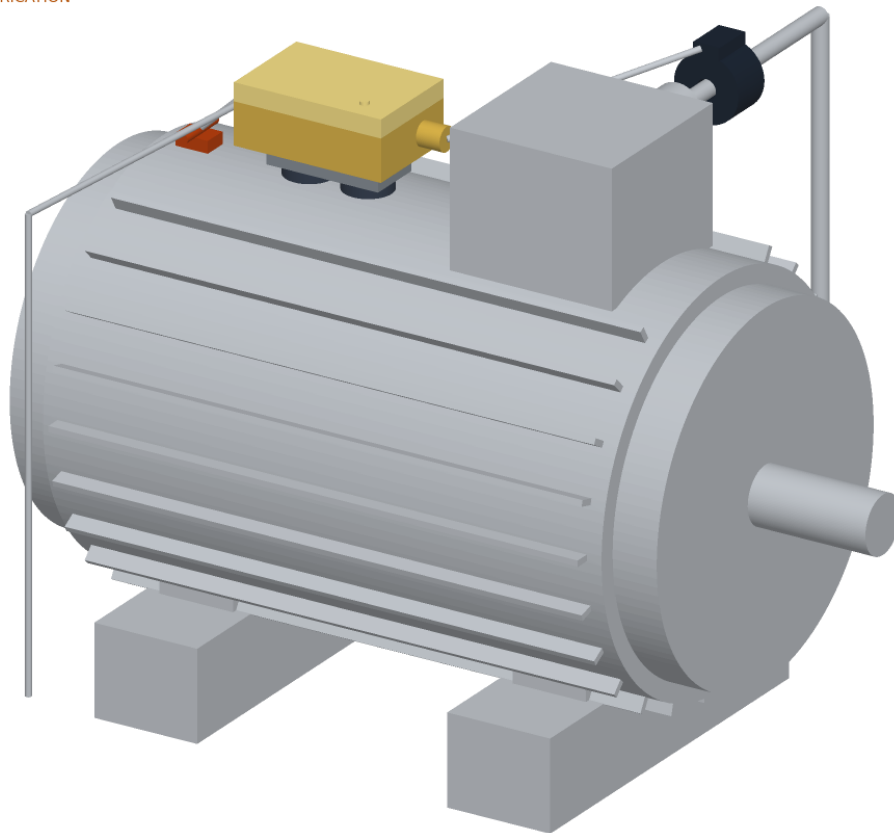
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■ Summary

MachinePulse is a small magnetic pod that clips onto an old machine's motor frame, plus a split-core current transformer (CT) on one phase conductor and a surface temperature probe. Every minute it reports run state, run time, load current, vibration velocity and frame temperature to a TwinKit gateway or any MQTT broker on the shop network, and every 15 minutes it sends a vibration spectrum. The TRL 3 calculation note MPL-CAL-001 finds that stock modules, a hand-made aluminium sensor block and a stock box meet twelve of the seventeen requirements for \$81.00 in parts, exactly the budget. One requirement is not met: fitting the CT without opening a live enclosure on many machines (R10). Four are at risk: current accuracy at the bottom of the CT range (R3), energy per shift (R4), the magnet mount's resonance inside the vibration band (R5) and the zero cost margin (R16). The design choices below were decided by Amish on 2026-09-25, going with the recommendations (MPL-DDR-001 and MPL-DDR-002). That decision added high-temperature pot magnets rated 120 °C, which clear hot frames to about 110 °C, and raised the budget from \$80 to \$81 to pay for them.

MachinePulse

CONCEPT, NOT FOR FABRICATION



Grey: 7.5 kW class induction motor and leads for scale

Figure 1. MachinePulse on a 7.5 kW class induction motor (grey, for scale), generated from the parametric model `cad/src/model.py` . Concept, not for fabrication.

■ How it works

1. **Sense current.** A voltage-output split-core CT (YHDC SCT-013 family, 5 to 60 A variants) clamps one insulated single-core phase conductor. The interface board biases its output at 1.50 V, so a full-range signal stays inside the ADC's linear window, and clamps the input against starting current. The pod samples at 2 kHz and computes RMS current every second.
2. **Sense vibration.** A wideband MEMS accelerometer (IIS3DWB class, dc to 6 kHz, 75 $\mu\text{g}/\text{VHz}$ noise density, 1.1 mA (ST)), set to ± 16 g, sits on a 20 mm round boss of an aluminium sensor block, directly above one of two pot magnets. The magnets pull the block onto the frame, so vibration reaches the sensor through metal, not through the plastic box.
3. **Sense temperature.** A DS18B20 probe in a stainless sleeve sits in a small magnetic clip with a thermal pad on the frame near a bearing.
4. **Summarize on the pod.** Every 60 s the ESP32-S3 controller records a 4 s vibration burst, decimates it from 26.7 kHz to 3.33 kHz as it reads the sensor, and computes velocity RMS over 10 to 1,000 Hz (the ISO 20816-1 band (ISO)), acceleration RMS, crest factor and the amplitudes at 1x and 2x running speed. It adds run state, run seconds, mean and peak current, starts, and temperature.
5. **Send and store.** The summary goes over Wi-Fi to an MQTT broker, by default on the TwinKit gateway. If the network is down, summaries queue in flash for about 14 days; spectra are kept only while space allows.
6. **Compare with the machine's own baseline.** During the first week of running, the gateway software learns a baseline per load band. Afterward it flags a change, for example vibration velocity above twice its baseline or temperature rise above baseline at the same load, for a person to inspect. Energy per shift uses a power factor curve built from the nameplate and the measured no-load current.

MachinePulse: data flow (estimates, MPL-CAL-001)
CONCEPT, NOT FOR FABRICATION



Figure 2. Data flow from machine to maintenance team. Values are estimates from MPL-CAL-001.

■ Main components

Table 1. Main components. Numbers match the BOM and Figure 3.

#	Component	Choice	Notes
1, 2	Enclosure	Stock IP54 ABS box, 100 x 68 x 40 mm, two M12 IP68 cable glands, LED window, 24 mm hole for the boss	Stock box keeps cost low (D7)
3	Controller	ESP32-S3 module board, 8 MB flash, no octal PSRAM, USB-C, Wi-Fi and BLE	Module rated to 85 °C; PSRAM variants are rated to 65 °C (D1)
4	Accelerometer	IIS3DWB-class 3-axis MEMS on an 18 x 18 mm adapter board, ± 16 g	ADXL345-class fallback misses R6 (D3)
5	Sensor block	76 x 36 x 10 mm aluminium with a 20 mm round boss 11 mm high through the box floor	Carries magnets and sensor; stiff path to the frame
6	Magnets	Two 32 mm high-temperature neodymium pot magnets, rated 120 °C, with M6 studs, 40 mm apart	Standard 80 °C grade would limit hot frames to about 81 °C; the 120 °C grade meets R12 (MPL-DDR-002, N1)
7	Interface board	Perfboard with 1.50 V CT bias and filter, series resistor and clamp diodes, 3.5 mm jack, probe connector, LED, button	No custom PCB for the first build
8	Current transformer	SCT-013 family, voltage output, 13 mm aperture, 5 to 60 A chosen per machine (30 A for the design case)	Voltage output has an internal burden, so it is never open-circuited (D4)
9	Temperature probe	DS18B20 in stainless sleeve, magnetic clip, thermal pad	
10	Power	Certified 5 V 1 A USB-C adapter and 2 m cable	Low voltage only in the pod (D2)
11	Steel pads	35 mm x 3 mm steel discs with epoxy, for aluminium frames	3 mm keeps about 75 % of the pull
12, 13	Hardware and stand-offs	Screws, silicone boot round the boss; four 6 x 5 mm nylon stand-offs	The stand-offs are the thermal break studied under D7

MachinePulse: exploded view

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- 1 Enclosure lid with LED window
- 2 Enclosure base with cable glands
- 3 Controller, ESP32-S3 module board
- 4 Vibration accelerometer, IIS3DWB class
- 5 Aluminium sensor block
- 6 Pot magnets, 32 mm, pair
- 7 Interface board (CT, probe, LED)
- 8 Split-core current transformer
- 9 Surface temperature probe
- 12 Silicone boot round the boss
- 13 Insulating stand-offs, nylon, four

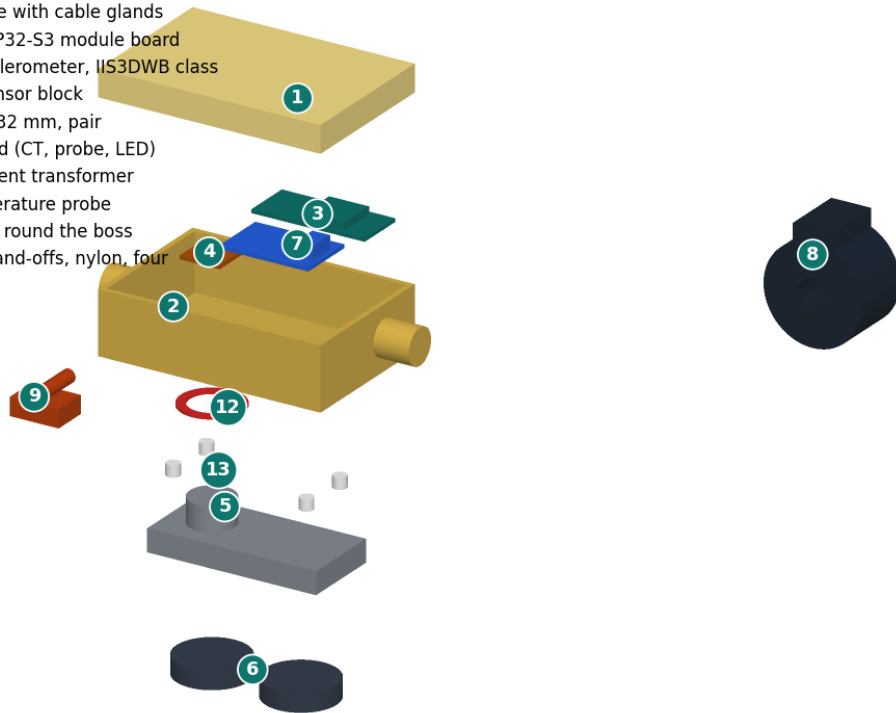


Figure 3. Exploded view with BOM numbers.

MachinePulse: cutaway

CONCEPT, NOT FOR FABRICATION

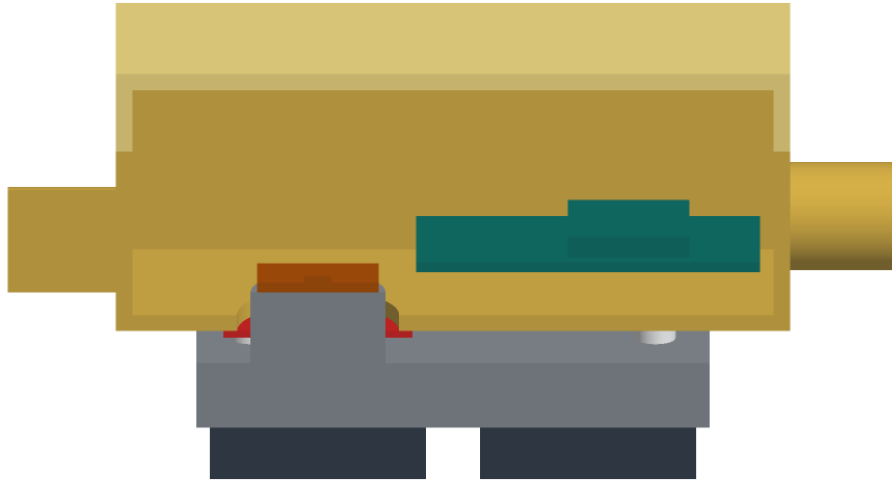


Figure 4. Cutaway through the accelerometer: the sensor sits on the boss of the aluminium block, which sits on the magnets; the box stands 5 mm clear of the block on nylon stand-offs and carries no vibration path.

The general arrangement drawing [MPL-DWG-001](#) (Rev P2, 1:1) gives the main dimensions: pod 100 x 68 x 63 mm high from the magnet face, 132 mm over the glands.

■ Key numbers

All values come from MPL-CAL-001 and are estimates for a paper proof of concept.

Table 2. Key numbers.

Quantity	Value	Requirement
Design motor full-load current	14.6 A, 49 % of a 30 A CT	Sizes the CT
Current accuracy	2.1 % RSS at the design point; 3.6 % RSS and 7.0 % worst case at 10 % of range	R3 at risk
Energy per shift	8.5 % RSS with a power factor curve; 26 % with a fixed power factor	R4 at risk
Velocity noise floor	0.037 mm/s RMS, 10 to 1,000 Hz	R6 met
Spectrum resolution	0.407 Hz bins (8,192 points at 3,333 Hz)	R7 met
Magnet mount resonance	About 927 Hz on a painted curved frame, 1,853 Hz flat	R5 at risk
Pod mass	About 0.30 kg	
Magnet holding	Slip margin 3.3 on a painted curved frame; pull-off 13.2	R11 met
Probe error	1.56 K at 80 °C, 1.69 K at 85 °C; 3.59 K at 100 °C	R8 met (restated target)
Hot frame, 40 °C air	Floor 53.9 °C and module about 58.9 °C at an 80 °C frame; accelerometer limit at a 109.6 °C frame, 120 °C magnets at 122.7 °C	R12 met
Power	0.42 W from 5 V; 0.61 W and 5.3 kWh a year at the wall	
Data	1.15 MB per day; worst delivery 66 s	R9 met
Offline store	13.9 days of summaries	R15 met
Parts cost	\$81.00 against \$81	R16 at risk

■ Key design choices

Decided by Amish, 2026-09-25: go with recommendation (MPL-DDR-001, MPL-DDR-002).

- **Stiff sensor path, plastic box for protection only (D7).** The accelerometer sits on the aluminium block's boss above a magnet, and the block sits on the magnets. The box rides on four nylon stand-offs 5 mm above the block, which keeps its floor 13 K cooler on a hot frame. The mount still resonates inside the upper vibration band on curved frames (R5).
- **Features on the pod, spectra on a slow schedule (D10).** Summaries every minute and a spectrum every 15 minutes come to about 1.15 MB a day, so one gateway can serve many machines. Raw bursts can be captured on demand for diagnosis.
- **One CT on one phase (D4).** Enough for run state, run hours and relative load, within budget. With a power factor curve the energy estimate is at risk rather than not met; a voltage reference or three CTs would cost \$10 to \$20 more.
- **Wi-Fi first (D1).** Small workshops usually have Wi-Fi, and 7.4 kB spectra are far too large for a LoRaWAN duty cycle. A summaries-only LoRaWAN variant could reuse the FieldNode radio core later.
- **High-temperature magnets (MPL-DDR-002, N1).** The magnets sit within about 2 K of the frame, so the standard 80 °C grade had no margin on an 80 °C frame. Pot magnets rated 120 °C cost about \$1.00 more for the pair and move the pod's hot-frame limit to about 110 °C, set by the accelerometer.

- **Mains-powered adapter, no battery (D2).** A certified 5 V adapter avoids lithium cells on a hot, vibrating frame.
- **Baseline, not fixed limits (D5).** Old machines differ too much for fixed alarm levels to be useful at first. Each machine is compared with its own first week, by load band, and flags go to a person.
- **Data stays local (D6).** MQTT to the TwinKit gateway, where the twin shows readings on the machine's model; any other broker works too.

■ Relation to other lab projects

- **TwinKit** is the default data home: its gateway accepts MQTT over Ethernet or Wi-Fi, stores data offline and shows readings against a model (TwinKit REQ R2). MachinePulse adds about 1.15 MB a day per machine, small against TwinKit's storage budget. MachinePulse is a natural first "small manufacturing" twin for it.
- **FieldNode** provides a LoRaWAN radio core if a summaries-only variant is wanted for sites without Wi-Fi. Only summaries would fit its 20-byte class payloads; spectra would not.
- **CalRig** can check the temperature probe against a reference before deployment. Vibration calibration needs a separate reference (a shaker or a known accelerometer) that CalRig does not provide.

■ Safety

Safety: Mains voltage. The CT goes around a live conductor. Fit it only with the machine isolated and locked off, only around a single insulated conductor, and have a qualified person open any terminal box or panel. Use only voltage-output CTs with an internal burden: a current-output CT left open-circuited on a live conductor can produce dangerous voltages ([OpenEnergyMonitor](#)).

Safety: Moving machinery. Fit and remove the pod and probe only when the machine is stopped and isolated. Mount on stationary frames, never on guards that open or on moving parts. Route leads away from belts, shafts, chucks and fans, and secure them with ties.

Safety: Hot surfaces and magnets. Motor frames and compressor heads can be hot enough to burn; the sensor block runs within a few kelvin of the frame. Strong neodymium magnets can pinch fingers and snap together; keep them away from pacemakers and magnetic media.

Safety: Power supply. Use only a certified 5 V adapter with the local safety mark. The pod contains no mains wiring and no lithium cells.

MachinePulse is a monitoring aid. It is not a protective device, it must never be wired into machine controls, and it does not replace overload relays, guards or scheduled maintenance.

■ Open questions

- R4 with one CT: accept "at risk", relax to a relative energy trend, or add a voltage reference. Awaiting Amish (MPL-DDR-001, O2).
- Co-design partner for alerts and dashboard. Awaiting Amish (MPL-DDR-001, O1).
- Measured values for the assumptions MPL-CAL-001 rests on: ESP32-S3 ADC residual error, magnet contact stiffness and pull on painted frames. Amish decided to keep the R3 and R5 targets and to measure the ADC residual and the mount resonance first, with a flat steel saddle for curved frames as the option to consider (MPL-DDR-002, N3). These need bench work, which belongs to TRL 4 and is on hold.

Concept media: [blueprint sheet](#), [interactive 3D model](#).